Building a World-Class ML Model for Dyslexic & ADHD Students in Literature

The Core Problem

A student with dyslexia opens *Things Fall Apart* by Chinua Achebe. They hit: *"He was tall and huge, and his bushy eyebrows and wide nose gave him a very severe look."* They freeze on "eyebrows," the letters swim, they lose the sentence, they lose the paragraph, they lose interest. A student with ADHD reads the same page three times and retains nothing because their attention drifted at line 2.

The model's job: keep that student in the story.

Deliverable 1: Real-Time Struggle Detection Engine

Forget quizzes after the fact. The model must detect struggle **as it happens**.

Signals to capture:

Signal	What it really means
Eye-tracking / gaze proxy (camera)	Which exact word caused a fixation or regression
Reading velocity changes	Sudden slowdown = decoding difficulty
Touch pressure (mobile)	Frustration correlates with harder taps
Pause duration between page turns	Processing overload vs. genuine engagement
Re-reading patterns	Scrolling back = lost comprehension
Time-of-day + session length	ADHD attention degrades predictably
Highlight/tap behavior	Student self-identifies difficult words

The ideal model: A **multimodal transformer** that fuses all these signals into a single "struggle score" updated every 2–3 seconds — not every page, not every chapter. Real-time.

Deliverable 2: Instant "What Does This Mean?" — Context-Aware Simplification

This is your highlight feature, but done properly:

Student highlights: *"The drums were still beating, persistent and unchanging. Their sound was no longer a separate thing from the living village."*

Bad simplification: "The drums kept playing. The sound was part of the village."
(Kills the literature. Kills the learning.)

World-class simplification:

> ***Simpler version:*** *"The drums kept beating steadily. After a while, their sound felt like it belonged to the village itself — like it was alive, just like the people."*
> ***Why Achebe wrote it this way:*** *He's showing that in Igbo culture, music isn't entertainment — it's part of life itself. The drums are the village.*
> ***Key word — "persistent":*** *means something that keeps going and won't stop. Think of a persistent alarm clock.*

What the model must do:

- Simplify **without destroying literary voice**
- Explain the **author's intent** (why it matters to the story)
- Give **vocabulary support** with relatable analogies
- Adjust the explanation level based on the student's **reading profile**
- Offer a **Kinyarwanda bridge** — connect the concept to something culturally familiar

Architecture: Fine-tuned LLM (distilled for edge deployment) with:

- Novel-specific context window (knows the full book)

- Student reading level embedding
- Literary device recognition (metaphor, irony, foreshadowing)

Deliverable 3: Adaptive Reading Mode Engine

Not one adaptation — a **dynamic combination** that shifts in real-time:

For Dyslexia:

Adaptation	When to trigger
Bionic Reading (bold first syllables)	Decoding slowdown detected
OpenDyslexic font switch	Persistent letter-reversal patterns
Syllable color-coding	Long word fixation (>3 seconds on a word)
Line spacing increase	Vertical tracking errors
Reading ruler (highlight current line)	Line-skipping detected
Audio narration of current sentence	Complete decoding breakdown

For ADHD:

Adaptation	When to trigger
Chapter chunking (break into 3-min segments)	Session start — always
Progress gamification ("5 paragraphs to next checkpoint!")	Attention drift detected
Inline comprehension micro-checks	Every 2–3 paragraphs
Ambient focus sounds	Student-selected preference
"Story so far" recap	Returning after a break
Cliffhanger breaks ("Want to know what Okonkwo does next?")	Natural story tension points

The RL policy network decides the combination and intensity. Not rules. Not if-then. A learned policy that knows *this particular student* at *this particular moment* needs *this particular cocktail* of adaptations.

Deliverable 4: Literature Comprehension Graph

A knowledge graph built per student, per book:

```
Student reads "Things Fall Apart"
    │
    ├── Characters understood: Okonkwo ✓, Ikemefuna ✓, Ezinma ~partial
    ├── Themes tracked: Masculinity ✓, Colonialism ✗ (not yet encountered)
    ├── Literary devices recognized: Proverbs ✓, Irony ✗
    ├── Vocabulary acquired: 47/120 target words
    ├── Chapters with >70% comprehension: 1,2,3,5
    ├── Chapters needing revisit: 4 (comprehension: 45%)
    └── Predicted struggle zones: Chapters 13–15 (vocabulary spike +
         narrative complexity increase)
```

Why this matters: The model doesn't just react — it **anticipates**. Before the student reaches Chapter 13, it pre-teaches vocabulary. It surfaces a "Previously in the story..." recap for complex plot threads.

Architecture: Graph neural network updated after every reading session, connected to a book-level difficulty map generated at content upload time.

Deliverable 5: The Student Reading Profile (Learner Embedding)

A **128-dimensional vector** that encodes everything about how this student reads:

- Decoding speed by word length
- Attention span curve (minutes before drift)
- Preferred learning modality (visual / auditory / read-write)
- Vocabulary level (per subject domain)
- Emotional response to difficulty (gives up fast vs. persists)
- Time-of-day performance patterns
- Which adaptations historically helped vs. annoyed them

This vector is the **input to every model decision**. It's what makes the system personalized, not generic. It updates every session. It transfers across books.

Deliverable 6: Teacher Intelligence Layer

Teachers don't want dashboards with graphs. They want answers:

> *"**Amina** is 2 chapters behind the class. She understands characters well but struggles with figurative language. **Recommended action:** Pair her with Jean for the proverbs discussion on Thursday."*

> *"**3 students** highlighted the same passage in Chapter 7. They may not understand the concept of tragic irony. **Suggested:** 5-minute class discussion before moving to Chapter 8."*

The model generates:

- Natural language student summaries (not charts)
- Specific, actionable teaching recommendations
- Class-wide pattern alerts (if many students struggle on the same thing)
- Predicted exam risk areas

Deliverable 7: Offline-First Inference Stack

Rwanda reality: intermittent power, shared devices, limited bandwidth.

Component	Spec
Core RL policy	< 20MB, runs on $50 Android phone
Simplification model	Distilled 1B-parameter LLM, < 500MB, quantized to INT4
Book content + difficulty maps	Pre-computed, cached per novel (~5MB/book)
Sync strategy	Delta-sync when online — only new session data uploads
Cold start	Works immediately with population-level defaults, personalizes within 3 sessions

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The Ideal Architecture (Summary)

```
┌──────────────────────────────────────────────────────┐
│                    STUDENT DEVICE                    │
│                                                      │
│  ┌──────────────┐   ┌──────────────────┐   ┌──────────────┐
```

```
│  │ Struggle  │──▶│  RL Policy   │──▶│Adaptation│ │
│  │ Detector  │   │  (PPO Agent) │   │ Renderer │ │
│  └──────────┘   └──────────────┘   └──────────┘ │
│       ▲                │                         │
│       │            ┌───▼────────┐                │
│  Raw signals       │  Student   │                │
│  (scroll, tap,     │ Embedding  │                │
│   pause, gaze)     │ (128-dim)  │                │
│                    └────────────┘                │
│                                                  │
│  ┌──────────────────────────────────────┐        │
│  │  On-Device Simplification LLM (distilled)│    │
│  │  + Book Context Window + Literary Device  │   │
│  │    Recognition + Kinyarwanda Bridge       │   │
│  └──────────────────────────────────────┘        │
└──────────────────────────────────────────────────┘
                    │ (when online)
            ┌───▼────────┐
            │ Cloud Sync  │
            │ + Retrain   │
            │ + Teacher   │
            │ Dashboard   │
            └────────────┘
```

What Makes This World-Class

1. **It preserves the literature.** Simplification doesn't dumb down — it bridges understanding while teaching the original language.
2. **It's invisible when working.** The best adaptation is the one the student doesn't notice. No pop-ups. No "you seem to be struggling!" Just quietly adjusting.
3. **It learns the individual, not the diagnosis.** Not all dyslexic students are the same. The embedding captures *this* student.
4. **It anticipates, not just reacts.** The comprehension graph predicts struggle before it happens.
5. **It works without internet.** Because Rwanda. Because reality.
6. **It gives teachers superpowers**, not homework. Actionable insights in plain language.